

The Derivative

Definition:

The derivative is the slope of the tangent line to the curve at any point x . It can also be defined as the rate of change at any point x .

The function must be defined at the point x .

Examples of where the function is **not** defined include:
Vertical Asymptotes (V. A.) and holes.

Examples of where the function is **not** differentiable are
sharp corners, endpoints, vertical tangents, *cusp*

1st Principles Definition of the Derivative

Lagrange Notation

read as "y prime"

$$y' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

read as "f prime of x" or

mt = $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Leibniz Notation

read as "dy by dx"

mt = $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Example 1: Determine y' if $y = \sqrt{x+1}$.

$$f(x) = \sqrt{x+1}$$

$$f'(3) = \frac{1}{4}$$

$$y' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$y' = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h+1} - \sqrt{x+1})}{h} \left(\frac{\sqrt{x+h+1} + \sqrt{x+1}}{\sqrt{x+h+1} + \sqrt{x+1}} \right)$$

$$y' = \lim_{h \rightarrow 0} \frac{\cancel{x+h+1} - (\cancel{x+1})}{h(\sqrt{x+h+1} + \sqrt{x+1})}$$

$$y' = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+1} + \sqrt{x+1}}$$

$$y' = \frac{1}{2\sqrt{x+1}}$$

$$\frac{dy}{dx} \bigg|_{x=3} = f'(3)$$

Example 2:

Determine $\frac{dy}{dx}$ at $x=3$ if $y = 2x^2 + 6x + 11$.

$$\frac{dy}{dx} \bigg|_{x=3}$$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)^2 + 6(x+h) + 11 - (2x^2 + 6x + 11)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + 6x + 6h + 11 - 2x^2 - 6x - 11}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{2x^2} + 4xh + 2h^2 + \cancel{6x} + 6h + \cancel{11} - \cancel{2x^2} - \cancel{6x} - \cancel{11}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 6h}{h} \therefore \frac{dy}{dx} \bigg|_{x=3} = 18$$

$$= 4x + 6 = 0 \therefore x = -\frac{3}{2}$$